

1. Yes I am using min heap for the implementation of Huffman tree. For Huffman tree we need to get 2 minimum elements and combine them. The fetching of minimum element is easier with min heap as the root node of min heap is always the smallest element. All operations in min heap take  $\log n$  time so it is faster than applying a whole search algorithm for minimum element in an array (which takes  $O(n)$  time).
2. original file size=10KB  
compressed file size=4KB
3. I have stored the meta data as a form of post order traversal of the Huffman tree. For every leaf node there is a 1 and the node data and for every intermediate node there is a 0. To reconstruct the tree during decompression I use a stack. For every 1 the node is appended into the stack and for every 0 the top 2 elements of stack are popped and combined to form an intermediate node which is reinserted in the stack. This process continues until only 1 element is left in the stack which becomes the root node.